Name	Subject Class	Class	Candidate Number
	2BIX01		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2018

BIOLOGY **8876/02**
17 AUGUST 2018
HIGHER 1 **2 hours**

Paper 2 Structured & Free-response Questions

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on the top of this page.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.
No additional materials are required.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6 or 7	
Total	**60**

For examiner's use

Section A

Answer **all** the questions in this section.

1 Cholesterol is synthesised in the smooth endoplasmic reticulum (SER) in liver cells by a series of enzyme-catalysed reactions. Cholesterol is then transported to the Golgi apparatus where they are packaged into vesicles and subsequently released into a membrane-bound duct of the liver.

Fig. 1.1 is an electron micrograph of a section of a liver tissue.

Fig. 1.1

(a) Name structure **T** in Fig. 1.1 and describe its role in liver cells.

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.... [2]

For examiner's use

(b) Both prokaryotes and structure **T** have membrane proteins to help them perform the role described in (a). Suggest how prokaryotes perform this role.

[3]

(c) Explain how prokaryotes illustrate the cell theory.

[3]

[Total: 8]

For examiner's use

2 Fig. 2.1 shows the stages involved in the synthesis of a protein in a eukaryotic cell.

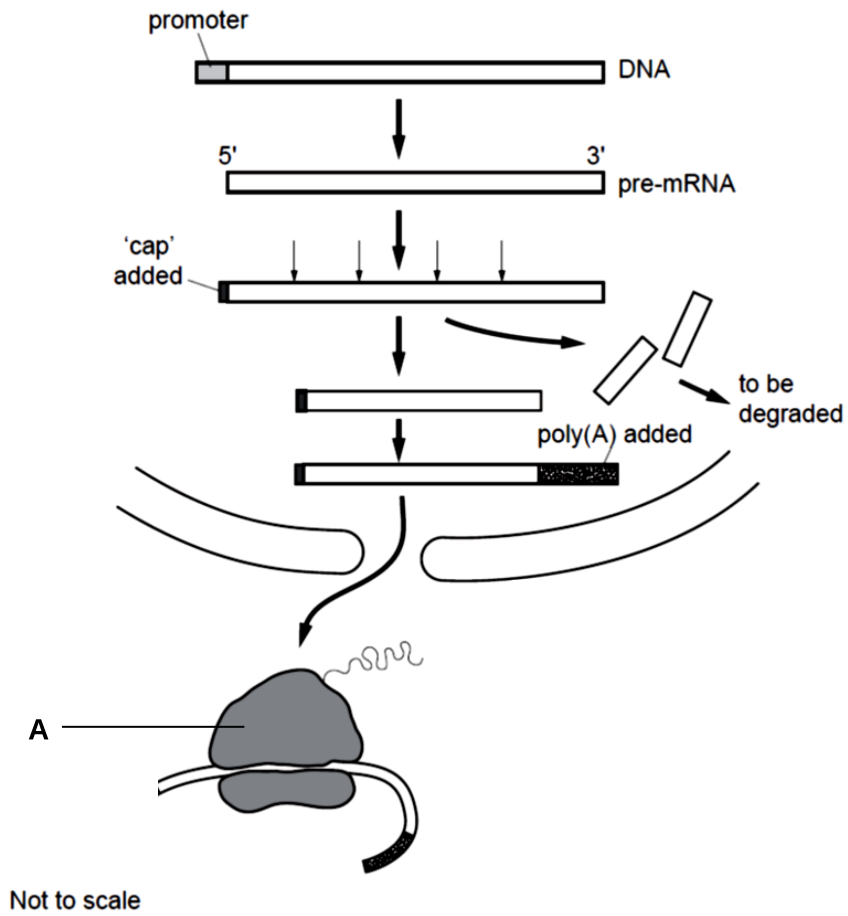

Fig. 2.1

(a) Describe the differences between the processes of protein synthesis in prokaryotes with that shown in Fig. 2.1.

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.... [2]

(b) Explain how the structure of organelle A is related to its function.

[4]

(c) Describe the formation of a bond found in both DNA & pre-mRNA.

[2]

[Total: 8]

For examiner's use

3 A study reported on the change in beetle sizes in a population due to climate change. Larger beetles are affected when compared to smaller beetles. This could be because larger beetles are less likely to get enough oxygen to sustain a higher metabolic rate caused by higher atmospheric temperatures. Fig. 3.1 shows the effect of a 2°C temperature rise on the size of larger and smaller beetles in a population over 100 generations.

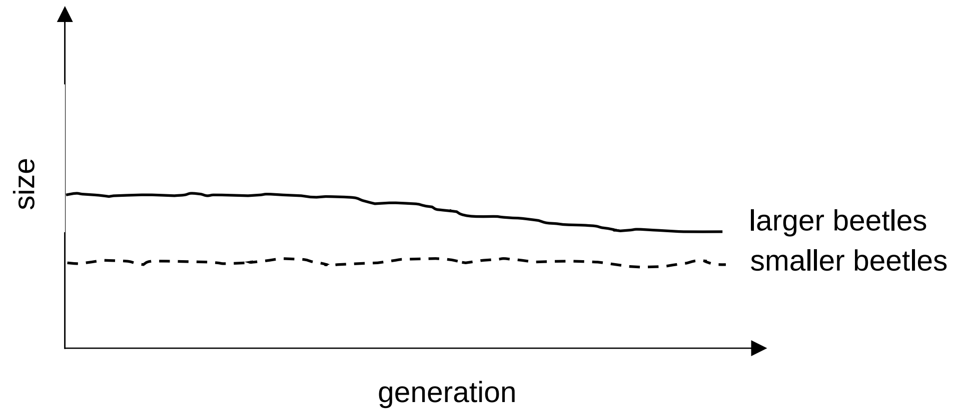

Fig. 3.1

(a) Define a 'population' of beetles.

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.... [2]

(b) Explain how a temperature rise of 2°C in may cause a change in the gene pool of a population of beetles.

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.... [4]

For examiner's use

(c) Comment on whether the alleles for large size in beetles will disappear from the population over time if temperature continues to increase as a result of climate change.

[2]

(d) Explain the consequences of a decrease in beetle size on the ecosystem.

[2]

[Total: 10]

For examiner's use

4 Radish can have two different shapes – elongated or round. The colour of radish can be red, white or purple. These two traits are determined by two genes found on different chromosomes.

When a true-breeding plant with elongated and red radish was cross-fertilised with a true-breeding plant with round and white radish, all the resulting plants have elongated and purple radish.

(a) (i) Using appropriate symbols, construct a genetic diagram to show the expected F2 phenotypic ratio when the plants with elongated and purple radish are self-fertilised.

[5]

For examiner's use

(ii) Each of the genes has only two allelic forms. Explain why the F2 phenotypic ratio does not follow the expected Mendelian ratio of 9:3:3:1.

[2]

(b) Suggest how the environment may affect the development of the colour of the radish.

[2]

[Total: 9]

For examiner's use

5 Table 5.1 shows the range of optimal temperatures for the growth of some major crops in the state of Iowa, USA. These crops are generally grown in April and harvested in October where average temperatures are between 16ºC and 28ºC.

Table 5.1

Type of crop	Temperature/ ºC		
	Optimal	Minimum	Maximum
Corn	22 - 25	20	32 - 34
Wheat	20 - 25	5	38
Soybean	25 - 28	10 -14	37 - 40

(a) **(i)** With reference to Table. 5.1 and your knowledge of enzymes, explain why Iowa is suitable for the growth of these crops.

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.... [2]

(ii) Suggest why corn productivity may be threatened by climate change.

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.... [3]

The level of H_2O_2 is high in crops which are under environmental stress. Catalase is an enzyme found in crops which protect the cells from oxidative damage by breaking down hydrogen peroxide (H_2O_2) into water and oxygen. Fig. 5.1 shows the effect of increasing H_2O_2 concentration on the rate of reaction of catalase.

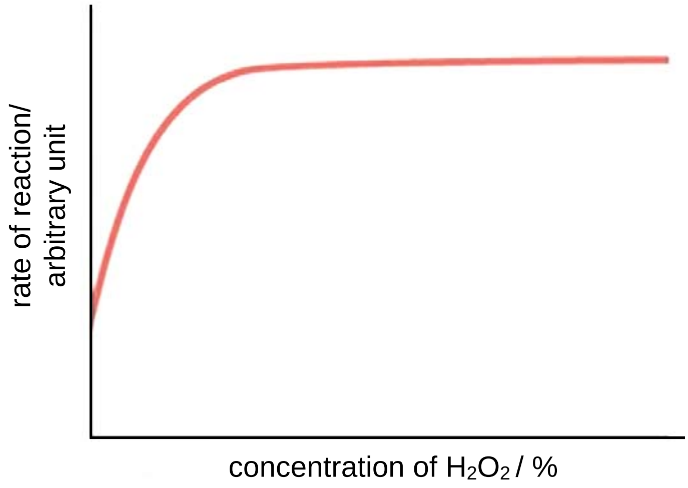

Fig. 5.1

(b) (i) With reference to Fig. 5.1, describe and explain the effect of increasing H_2O_2 concentration on the rate of reaction of catalase.

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.... [2]

(ii) With reference to the information provided, suggest why the cells in soybean suffer from oxidative damage when the temperature reaches 38ºC.

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.... [3]

[Total: 10]

For examiner's use

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

You answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)** and **(b)**, as indicated in the question.

6 **(a)** Outline the processes resulting in chromosomal aberrations. [6]

(b) Discuss how the structures of biomolecules can account for their roles in plants. [9]

[Total: 15]

7 **(a)** Discuss the significance of the movement of substances across membranes to photosynthesis. [6]

(b) Outline the key features of stem cells which make them useful for research and medical applications, and discuss the ethical implications which may arise from these applications. [9]

[Total: 15]

For examiner's use

For examiner's use

For examiner's use

For examiner's use